Academic Year: 2026-2027
Semester: III
Class / Branch: SE IT
Subject: ADSAL LAB
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Date of Performance:13/10/2026
Date of Submission:13/10/2026

Experiment No.12

Aim: Implement Longest Common Subsequence (LCS) algorithm.

Code:

```
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
return (a > b) ? a : b;
}

int lcs(char X[], char Y[]) {
int m = strlen(X);
int n = strlen(Y);
int L[m + 1][n + 1];

for (int i = 0; i <= m; i++) {
for (int j = 0; j <= n; j++) {

if (i == 0 || j == 0)
L[i][j] = 0;

else if (X[i - 1] == Y[j - 1])
L[i][j] = L[i - 1][j - 1] + 1;

else
L[i][j] = max(L[i - 1][j],
L[i][j - 1]);

}
}
```

```
return L[m][n];
}

int main() {
char X[100], Y[100];

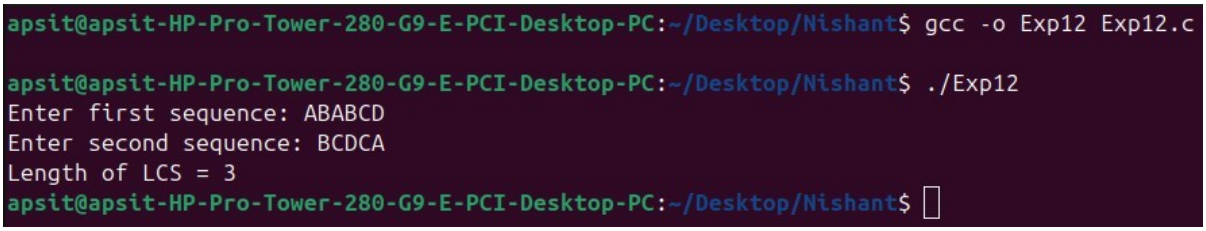
printf("Enter first sequence: ");
scanf("%99s", X);

printf("Enter second sequence: ");
scanf("%99s", Y);

printf("Length of LCS = %d\n", lcs(X, Y));

return 0;
}
```

Output Screen Shots:



Conclusion:

The Longest Common Subsequence (LCS) algorithm was successfully implemented using Dynamic Programming. The program compares the two sequences and stores the results of smaller subproblems in a DP table to efficiently find the LCS length. For the given example, "ABCBDAB" and "BDCABA", the length of the LCS is 4. This experiment helped in understanding how Dynamic Programming can be used to solve problems efficiently by avoiding repeated calculations.